International Journal of Hydrogen Energy 250 (2026) 156060

Contents lists available at ScienceDirect

International Journal of Hydrogen Energy

journal homepage: www.elsevier.com/locate/he

Spatially resolved electrochemical characterization of a PEM fuel cell stack during durability testing

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ARTICLE INFO

Keywords:
Proton exchange membrane fuel cell
Electrochemical impedance spectroscopy
Current density distribution
Spatially resolved characterization
Durability test
Aging effects

ABSTRACT

The durability of proton exchange membrane fuel cells (PEMFC) is a decisive factor for their use in heavy-duty applications. In order to increase durability, it is important to identify and understand the aging mechanisms. In large-area PEMFCs, inhomogeneities occur across the active area right from beginning of life, which have a decisive influence on degradation. This paper presents results on aging phenomena with a particular focus on spatially resolved degradation. A heavy-duty relevant load cycle was applied for more than 2000 h to a PEMFC stack with an active area of 280 cm^2. At regular intervals, detailed electrochemical characterizations such as polarization curves, sensitivity tests, and electrochemical impedance spectroscopy at cell level and spatially resolved were performed. Degradation impact increases with decreasing pressure, especially in the resistance of the oxygen reduction reaction. Mass transport losses are primarily affected in the cathode outlet area and show the greatest dependence on stoichiometry.

1. Introduction

Fuel cells are a climate-friendly alternative to combustion engines and can play an important role in the decarbonization of the transport sector. Compared to battery-electric systems, the proton exchange membrane fuel cells (PEMFC) offer advantages in terms of range and short refueling times, which makes them particularly attractive for heavy-duty transport applications [1,2]. For competitive deployment, the durability of the fuel cell stack is a decisive factor. To assess durability and efficiency, the investigation of the membrane electrode assembly (MEA) is of particular importance since they lead fuel cell performance degradation during aging. In addition, heterogeneities across the active area already observed at the beginning of life (BoL), intensify during operation. These local differences in degradation significantly affect overall performance and represent a major challenge for long-term stability [3].

Among degradation mechanisms, membrane degradation is considered the most critical, as it can ultimately lead to cell failure, whereas degradation of other components generally results in gradual performance losses [4]. Membrane failure is mainly caused by the formation of cracks or pinholes, which can induce excessive gas crossover and prevent safe operation. Such defects may already occur during manufacturing or assembly and may be promoted by mechanical stresses arising from hydration and dehydration cycles, for example during start-up and shutdown [5,6]. Other membrane-related aging phenomena, such as membrane thinning and chemical decomposition, primarily lead to performance losses. These processes are typically associated with dynamic operating conditions involving dynamic cycles in potentials and humidity levels. Another mechanism is the detachment of the catalyst layer from the membrane [7], creating cavities that facilitate local water accumulation, increase charge transfer resistance, and hence reduce cell performance. Catalyst aging is causing the loss of

This article is part of a special issue entitled: EFCHC25 conference (Iulianelli) published in International Journal of Hydrogen Energy.

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https://doi.org/10.1016/j.ijhydene.2026.156060
Received 5 March 2026; Received in revised form 21 April 2026; Accepted 12 June 2026
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electrochemically active surface area (ECSA), which directly results in voltage degradation [6,8]. Several mechanisms occur, including platinum particles dissolution, agglomeration and electrochemical Ostwald ripening. The two latter lead to larger particles with a reduced surface-to-volume ratio, while dissolved platinum cations may migrate and be reduced as non-active particles into the membrane. A comprehensive overview of PEMFC aging mechanisms is provided in Ref. [6].

The aim of this work is to achieve a deeper understanding of aging phenomena in a metallic PEMFC stack. To this end, a heavy-duty relevant durability test is conducted. The stack performance is evaluated at regular intervals using electrochemical characterization techniques, allowing the quantification of key parameters contributing to degradation. The focus is on spatially resolved aging effects across the active area. Local variations in current density are assessed using a current density distribution measurement board (CDDM). Previous studies considering spatially resolved current density distributions were conducted on the development of accelerated durability test (ADT) by Koprek et al. [9] and Schüttoff et al. [10]. Further analyses of local current density evolution with increasing degradation can be found in Refs. [11,12], while the influence of operating conditions is discussed in Refs. [13–17].

Electrochemical impedance spectroscopy (EIS) is a powerful method for analyzing resistances of transport, conduction, and reaction processes occurring at different frequencies. By separating individual loss mechanisms, EIS enables a more detailed interpretation of aging-related changes. Cell-level EIS measurements are well established in PEMFC aging research. Chandran et al. [7] used EIS to investigate the influence of different load profiles on degradation. EIS is widely applied in ADTs and accelerated stress tests (ASTs) [9,10,18,19]. While cell-level EIS measurements provide valuable insight into global processes, they do not capture effects caused by non-uniform local conditions, which strongly influence local performance and degradation. By using additional expansion boards for the CDDM, spatially resolved EIS measurements across the active area become possible. Although this technique is not yet common, it has been found promising by several studies [20–25]. Most previous works have focused on the influence of flow field design [22,23,26] or water balance [21,24,25], and predominantly on single cells with relatively small active areas. However, for heavy-duty applications, investigating larger active area is more relevant as generally leading to stronger spatial heterogeneities that significantly affect performance and aging behavior. So far only very few studies address large-format cells or stacks. One notable example is the work of Haimerl et al. [12], who investigated spatially resolved impedance behavior of a PEMFC with an active area of 285 cm^2 under various operating conditions and during freeze-start degradation.

Thus, an aging study focusing on the investigation of application relevant cell-sizes in a short-stack employing a combination of advanced analysis methods including spatially resolved EIS appears most relevant. Building on our previous investigations of BoL performance phenomena [27], the present study extends this work by providing a systematic analysis of spatially resolved aging effects in a PEMFC with an active area of 280 cm^2. In contrast to prior studies as [7,10,22–24,28], which predominantly focus either on global aging phenomena or localized analyses at small cell level, this work combines spatially resolved electrochemical impedance spectroscopy and current density measurements within a long-term durability test conducted under realistic operating conditions. This approach enables the direct correlation of local electrochemical properties with degradation behavior during operation. In particular, the influence of different operating conditions on the evolution of inhomogeneities is analyzed, allowing for a more detailed assessment of local degradation mechanisms across the active area. By applying spatially resolved diagnostics to a large-area cell format, this study contributes to closing the research gap between fundamental investigations employing small, typical laboratory-scale single cells and application-relevant PEM fuel cell stacks, especially in the context of heavy-duty applications.

2. Material and methods

2.1. Test-setup

The tests were conducted as part of the RealHyFC project [29]. The generic HyFaB stack [30] with an active area of 280 cm^2 and a project-specific MEA served as the stack hardware using: a 10-cell stack for the durability test (see Section 2.3), and a 8-cell stack for the BoL EIS sensitivity analysis. The BoL sensitivity study was conducted on a separate stack, as this module was deliberately removed from the durability test to ensure that any initial degradation could be solely attributed to the applied load cycle. To enable a comparison between the sensitivity behavior of the different aging states, the comprehensive sensitivity study, including EIS analysis during sensitivity measurements, was therefore repeated on a second, fresh stack. The number of cells for the sensitivity stack was reduced due to test bench limitations, as the 10-cell configuration did not allow the highest stoichiometry to be tested at the highest load point. It should be noted that identical MEAs from the same production batch were employed for both stacks. Both stacks integrated in the middle of the cell package a CDDM from S++ Simulation Services with a resolution of 29x23 corresponding to 667 measurement segments for the local current density. In the 10-cell stack, the CDDM was integrated between cells 5 and 6, whereas in the 8-cell stack it was positioned between cells 4 and 5. In both cases, conventional bipolar plates of the adjacent cells were replaced by end bipolar plates, and additional gas diffusion layers were inserted on both sides of the CDDM. Consequently, as usual with the method, the measured local current density represents the combined contribution of several cells. The individual segments of the current density measurement board are calibrated by the manufacturer using a two-point calibration method. The measurement uncertainty of the individual measurement segments is not specified in detail. According to the manufacturer, however, the expected measurement error is at least one order of magnitude smaller than the measurement error caused by lateral cross-conductance within the cell. Thus, the contribution of sensor calibration to the total uncertainty of the spatially resolved current density measurement can be considered negligible compared to blurring mainly induced by the distance between the catalyst layer and the measurement board and in-plane conductivity. The latter effects can hardly be quantified as there is no established measurement technique for local current density within the catalyst layer. The stacks were operated in counterflow mode, with the coolant flowing in the same direction as the air flow. The tests were performed on a Greenlight G200 test bench. Further details on the test setup, including current density correction factors and the break-in procedure, are given in Ref. [27].

2.2. Characterization methods

Details on the characterization methods used can be found in Ref. [27]. For the purpose of completeness, they are briefly described below.

2.2.1. Polarization curves

Six different polarization curves were measured. However, only the polarization curve under nominal RealHyFC conditions is relevant for the purpose of this paper. These conditions applied for the durability test and for the other characterization methods are shown in Table 1.

The coolant flow was kept constant and selected so that the temperature difference between the inlet and outlet was 10 K at the highest load point of 1.8 A cm^{-2}. A total of 16 different load points in a range from 0 A cm^{-2} to 1.8 A cm^{-2} were measured in both descending and ascending order. For the durability test the minimum current for stoichiometric operation was set to 0.3 A cm^{-2}, while 0.5 A cm^{-2} was used for the full sensitivity test.

Table 1
Nominal RealHyFC operating conditions applied for the polarization curve, the EIS measurements and the durability test.

	Inlet temperature /°C	Humidity /%	**Outlet Pressure /kPaG**	**Stoichiometry /-**	**Gas composition**
Anode	75	50	160	1.5	80 % H_2/20 % N_2
Cathode	75	35	150	1.7	100 % Air
Coolant	70	-	-	-	-

2.2.2. Sensitivity test

Sensitivity tests were performed to investigate the influence of operating conditions on performance. A partial sensitivity test, which includes fewer measurement points, and a full sensitivity test were performed. Both sensitivity tests studied the influence of temperature, humidity, pressure, and stoichiometry at four different current densities (cf. Table 2). The parameters were changed together for both anode and cathode. The selected operating parameter ranges were defined based on the nominal RealHyFC conditions and systematically varied towards both higher and lower values within the ranges typical for the operation of PEMFCs. The variation limits take into account the constraints imposed by the stack hardware, the MEA components and the test bench. Furthermore, contributions and advice of the industrial partners within the RealHyFC consortium were reflected to maximize the relevance of the chosen parameters. The partial sensitivity test was performed every 50 cycles during the durability test as follows: only one parameter is varied at a time, while the others remain constant at the thick, underlined nominal values. This results in a total of 13 different sets of operating conditions for the partial sensitivity test. For the full sensitivity test, used for the BoL test and at the end of the durability test, all possible 240 conditions were tested ending with 960 measurement points.

Details regarding current density ramp, dwell times and conditioning times for the sensitivity test and the polarization curve can be found in Table 3.

2.2.3. Electrochemical impedance spectroscopy

The EIS measurements were performed using a Zahner potentiostat of type Zennium X in combination with four additional 1 V-PAD4 cards [31] and an external electrical load of type EL1002 [32]. The additional PAD4 cards enable the measurement of 16 impedance spectra in parallel with a reference impedance. Both the cell impedance and the spatially resolved impedance were measured galvanostatically. For the cell impedance, the alternating current (AC) was 5 % of the direct current (DC). The spatially resolved measurements showed that a slightly higher alternating current results in less noise in the measurements. Accordingly, an AC of 7 % of the DC was used for the spatially resolved measurements. The spatially resolved EIS measurement was performed using the CDDM and special expansion boards that transmit the current from the measurement segments to the Zahner device. It is assumed that the metallic bipolar plate represents an equipotential surface and thus the voltage is constant over the entire active area. For the EIS measurement, the voltage of a reference cell and the AC recorded by the CDDM are used. The cell directly above the CDDM served as the reference. For the spatially resolved measurement, the CDDM was divided into 16 different zones based on the current density distribution map observed at 1.8 A cm^{-2}, with 7 zones selected in the cathode inlet area (blue-based colors), two in the middle area (yellow-based colors), and 7 in the cathode outlet area (green-based colors) as shown in Fig. 1.

Table 2
Operating conditions of the sensitivity tests. For pressure, humidity, and stoichiometry, anode and cathode (A/C) conditions are changed simultaneously. Nominal conditions shown in bold and underlined.

Parameter/Variation	1	2	3	4	5
Temperature/°C (coolant inlet)	55	60	65	**70**	75
Pressure A/C/kPaG (outlet)	60/50	110/100	**160/150**	210/200	
Humidity A/C/%	30/30	**50/35**	70/60	90/90	
Stoichiometry A/C/-	1.4/1.6	**1.5/1.7**	1.9/2.1		
Current density/A cm^{-2}	0.36	0.9	1.8	2.3	

Table 3
Current density ramps, dwell times and conditioning times for the characterization methods.

	Polarization curve	**Sensitivity test**
Current density ramps	2 A s^{-1}	2 A s^{-1}
Dwell time per load point	5 min for $j \geq 0.3$ A cm^{-2} 2 min for $j < 0.3$ A cm^{-2}	5 min
Conditioning time	30 min at 1 A cm^{-2}	Change of temperature or humidity: 30 min at 1 A cm^{-2} Change of pressure or stoichiometry: 10 min at 1 A cm^{-2}

The uncertainty of the impedance measurement was estimated based on the specified measurement uncertainties of the PAD4 cards and the EL1002 module. The combined uncertainties of voltage U and current I were applied to the impedance Z

$$Z = \frac{U}{I} \tag{1}$$

using the Gaussian error propagation. For a load point of 1.8 A cm^{-2}, the relative uncertainty of the impedance magnitude covered by this estimation is below 0.3 %, indicating a very low contribution of this measurement uncertainty to the overall signal. It should be noted that additional uncertainties arising from cable-related effects, such as contact resistances, inductive contributions, and parasitic coupling, are not explicitly included in this estimation. However, these effects are expected to be small compared to the intrinsic measurement uncertainties of the instrumentation, i.e. the Zahner device for cell impedance and additionally the effects related to the CDDM for the spatially resolved EIS. Thus, the observed variations in the impedance can be attributed primarily to physical effects rather than measurement artifacts and do not significantly affect the overall interpretation of the results.

2.3. Durability test procedure

To investigate the effects of aging, a durability test was carried out applying a load cycle defined within the RealHyFC project and based on the PEMTASTIC load cycle [33]. The cycle has a duration of approximately 12 h and includes three short and one longer stop. The time course of the load cycle (left) and the percentage distribution of the load points in the cycle (right) are shown in Fig. 2. The operating conditions correspond to the nominal RealHyFC conditions.

A schematic representation of the test procedure for the overall test procedure applied for the durability test is shown in Fig. 3.

The break-in aiming to achieve the full performance of the cells, is followed by the full characterization sequence, including the methods described before: six different polarization curves, the partial sensitivity test, EIS at cell level and spatially resolved, as well as cyclic voltammetry

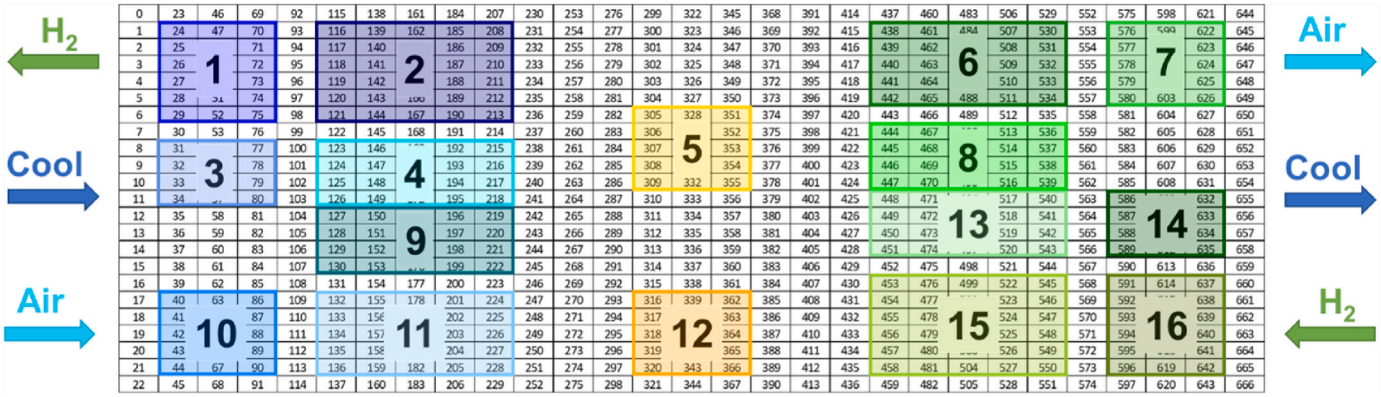
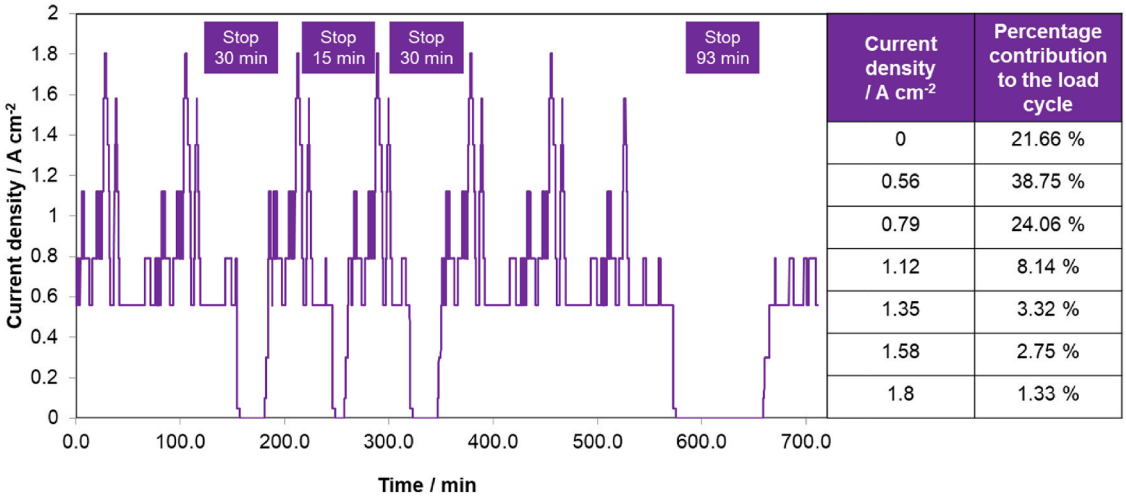


Fig. 1. Division of the CDDM measuring segments into a total of 16 different zones across the active area of the fuel cell.



Current density / A cm^{-2}	Percentage contribution to the load cycle
0	21.66 %
0.56	38.75 %
0.79	24.06 %
1.12	8.14 %
1.35	3.32 %
1.58	2.75 %
1.8	1.33 %

Fig. 2. Current density profile (left) and percentage contribution of load points (right) of the load cycle from the RealHyFC project, which is based on the load cycle of the PEMTASTIC project [33]. Current range is defined by the specifications of the stack.

(CV) and linear sweep voltammetry (LSV). The voltammetry measurements are not discussed in detail here, as the results are not relevant for this work focused on spatially resolved data.

After every 25 load cycles, a partial characterization is performed, consisting only of the six polarization curves, and after every 50 cycles, the stack undergoes the full characterization. A passive air/air start was integrated into the characterization sequence after the first 50 cycles, followed by an additional nominal RealHyFC polarization curve to measure reversible voltage losses. During passive air/air start, air diffuses to the anode side, allowing to regenerate reversible voltage losses due to temporary concentration gradients or water accumulation. As air/air starts could promote some degradation mechanisms [18,34], especially in the catalyst layer, they are only performed as part of the characterization and in limited numbers. As the goal of this study was to complete more than 2000 operating hours (corresponding to about 170 cycles), the durability test was ended after a relevant number of 200 cycles, to perform a full characterization. Subsequently, the final characterization including spatially resolved EIS measurements and cell EIS measurements of the reference cell under the partial sensitivity test conditions is carried out, followed by the full sensitivity test and a final full characterization excluding the partial sensitivity test.

3. Results and discussion

The results of BoL nominal polarization curve, partial sensitivity test, cell EIS and spatially resolved EIS under nominal conditions have already been described in detail in Ref. [27]. The following section takes a closer look at aging and analysis of the EIS measurements during the sensitivity test.

3.1. Voltage progression in polarization curves and sensitivity tests

The progression of the nominal polarization curves recorded immediately following the load cycle is shown in Fig. 4. It is clear that the cell voltage decreases over time. The largest jump is between the BoL curve and the curve after 300 h (25 cycles). In this initial period, the voltage loss is about half the total, with a degradation rate of 63.3 μV h^{-1} at 1.8 A cm^{-2}. Looking at the entire durability test, the total degradation rate is 15.7 μV h^{-1}. However, this value includes both reversible and irreversible losses. If the nominal polarization curves after air-air start are considered, excluding the reversible cell voltages, the degradation rate decreases to 13.0 μV h^{-1} at 1.8 A cm^{-2}. This result is consistent with an initial degradation known to be the most severe, as a significant loss of the ECSA occurs at the beginning due to catalyst aging, while afterwards the degradation rate stabilizes [8,11]. Looking at the degradation rate after 1500 h, it is only 3.8 μV h^{-1}. The observed degradation rates are in good agreement with degradation rates reported in the literature for PEM fuel cells [9,35–38]. It can be noted that the hysteresis between increasing and decreasing current, particularly visible below 0.5 A cm^{-2} is not affected by aging, showing a limited impact of aging on this feature associated with global sensitivity to humidity at least on the average performance.

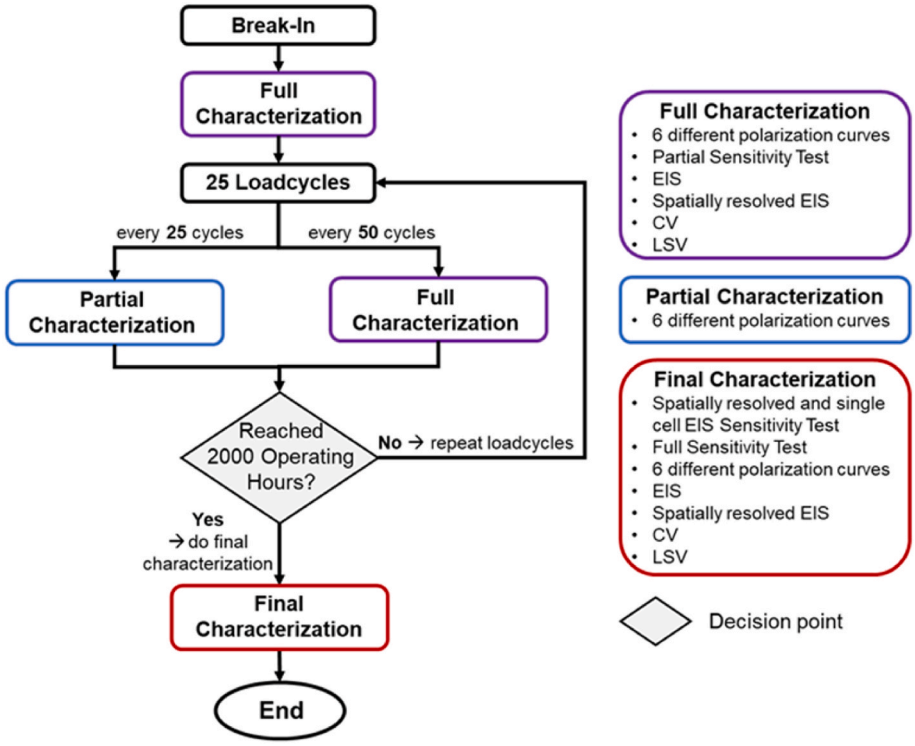


Fig. 3. Schematic representation of the durability test procedure with a description of the characterization sequences. For the purpose of our study, an operating hour is defined as the time under load.

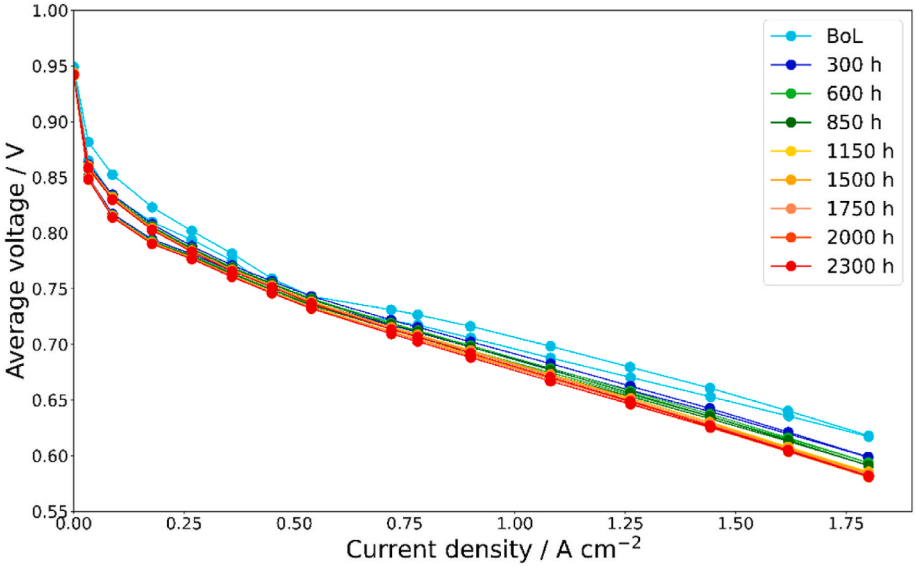


Fig. 4. Nominal RealHyFC polarization curve from the characterization sequences of the durability test including reversible and irreversible cell voltage losses.

The partial sensitivity test shows the change in cell voltages under various operating conditions. The results of the partial sensitivity test for BoL and after completing all 200 cycles are shown in Fig. 5.

The cell voltage loss is clearly visible in all four variations, and increasing with the current. For the temperature, humidity, and stoichiometry variations, the voltage losses are approximately the same under all operating conditions (about 50 mV at the highest current), and consistent in values with the nominal polarization curves, not showing strong influence of aging on the sensitivity. In contrast, the influence of aging on PEMFC sensitivity becomes evident under varying operating pressures, with much more pronounced performance losses as the pressure decreases. It appears that the aging-related performance becomes more sensitive to reduced reactant partial pressures and present increased susceptibility to mass-transport limitations. Increasing operating pressures enhances reactant availability and cell operation, as for the non-aged case, but with a much more beneficial effect as seems required after aging. The higher partial pressure mitigates the impact of aging by improving oxygen transport to the catalyst layer, thereby supporting more uniform and robust cell performance.

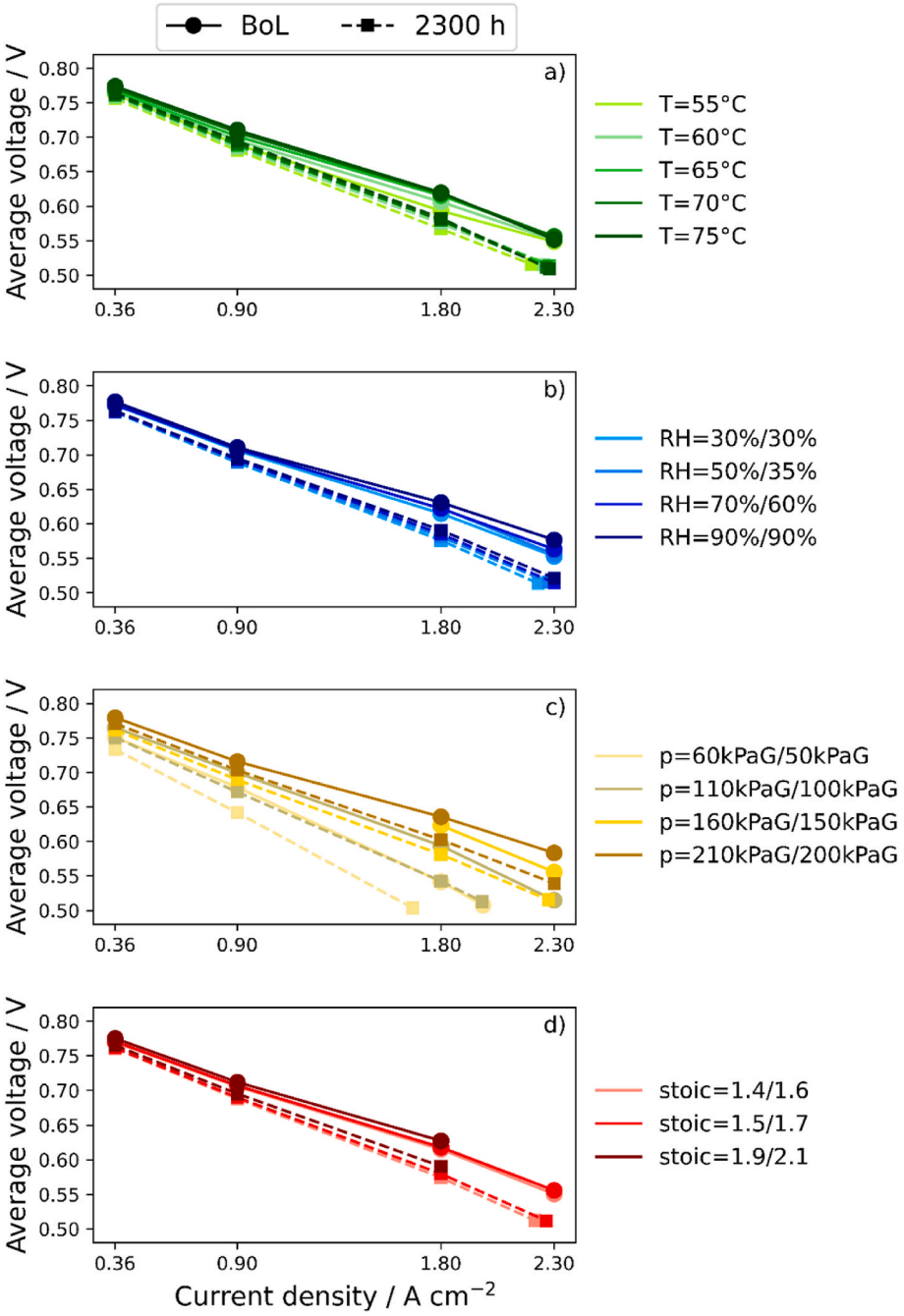


Fig. 5. Partial sensitivity test at BoL and after 200 cycles (2300 h), a) temperature variation, b) humidity variation, c) pressure variation, d) stoichiometry variation.

3.2. Current density distribution analysis

The current density distribution changes with aging. The current density distribution during the descending branch of the nominal RealHyFC characteristic curve after air/air start is shown in Fig. 6 for three load points over the course of aging. For the line-graphs, the active area is divided into 27 columns in the flow direction according to the current density measurement segments, and the mean value is calculated within each column. To illustrate the systematic variation in current density perpendicular to the channel, the BoL curve at 1.8 A cm^{-2} shows the minimum and maximum current densities within the respective columns along the cathode flow direction as error bars.

The 2D images at BoL and after 2300 operating hours show a typical current density distribution pattern, with lower values in the inlet and outlet regions as well as at the edges, and higher values in the central region of the active area. A comparison of the current density over time reveals a visible shift in the current density maximum towards the cathode inlet. Furthermore, the inhomogeneity increases over time and the current density maximum reaches higher values after 2300 h. This is confirmed by the 1D plots and agrees well with previously reported results in the literature [37,39]. During the first 600 h of aging, a significant shift in current density can be observed, which thereafter changes only very slightly, as seen for the polarization curves. However, the overall shape does not change, only the low local current densities at air inlet are becoming even lower but only for the lowest load, compensated by higher values in the middle area from 5 to 14. For the higher loads, no effect noticed at air inlet but an increased slope between the maximum area, always nearby the air inlet, and the area from middle to outlet. Aging has thus affected the operation at air inlet at low current, probably caused by a higher susceptibility to dry conditions. This is also

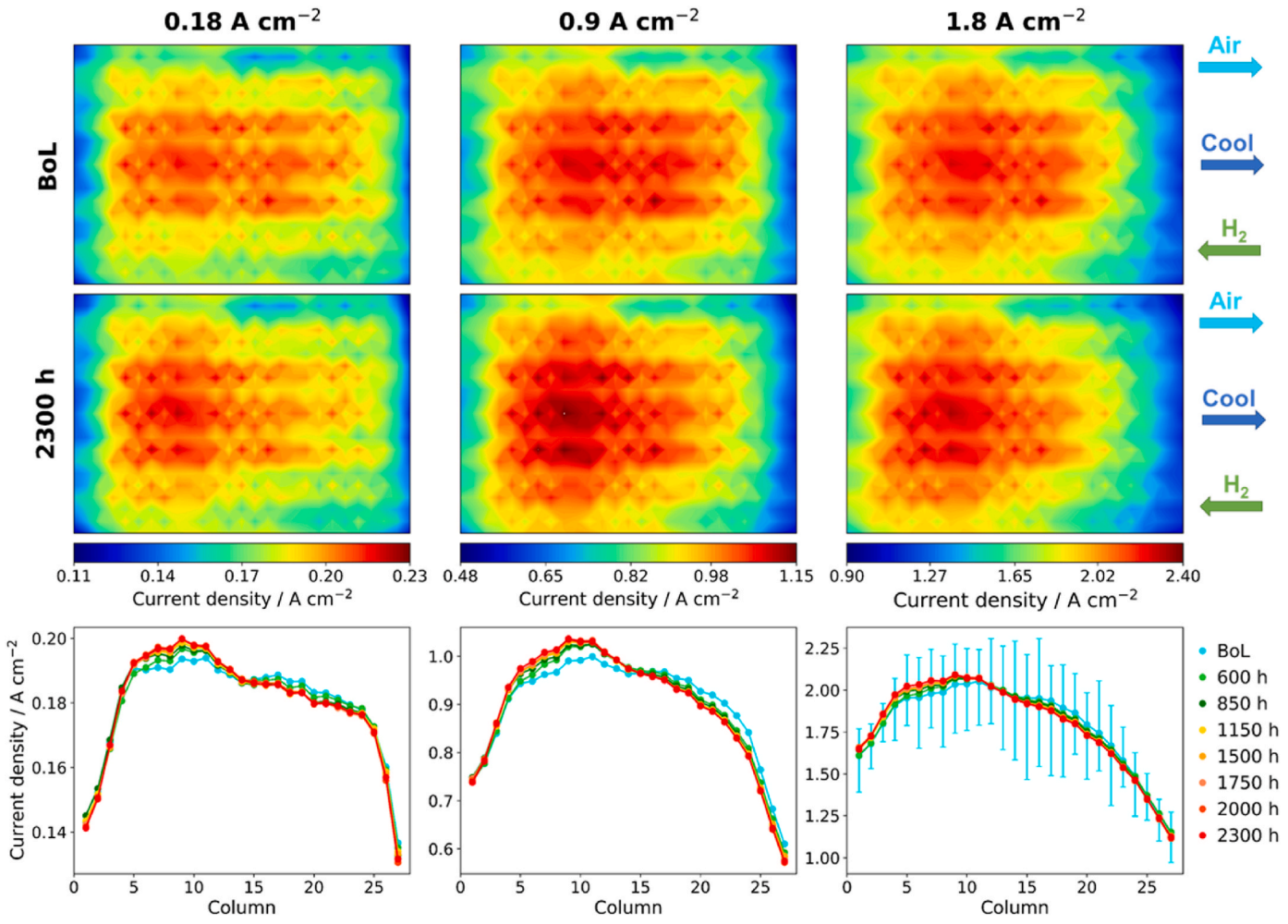


Fig. 6. Current density distribution during the polarization curve under nominal RealHyFC conditions after air/air start at 0.18 A cm^{-2}, 0.9 A cm^{-2} and 1.8 A cm^{-2}. Representation of the current density distribution at BoL and after 2300 h as a 2D graph, as well as the evolution of the current density over several aging time points as a 1D graph. Indicating the systematic distribution of current density with error bars showing the minimum and maximum values using the BoL curve at 1.8 A cm^{-2}.

confirmed with small but visible continuous decrease of current densities for the very first column.

With the help of spatially resolved polarization curves, the average current density of the 16 defined individual zones (cf. Fig. 1) can be plotted against the average stack voltage (cf. Fig. 7).

The spatially resolved polarization curves demonstrate that the current density varies greatly between zones. The spread of local current densities is analyzed at the highest load point (about 0.6 V). The

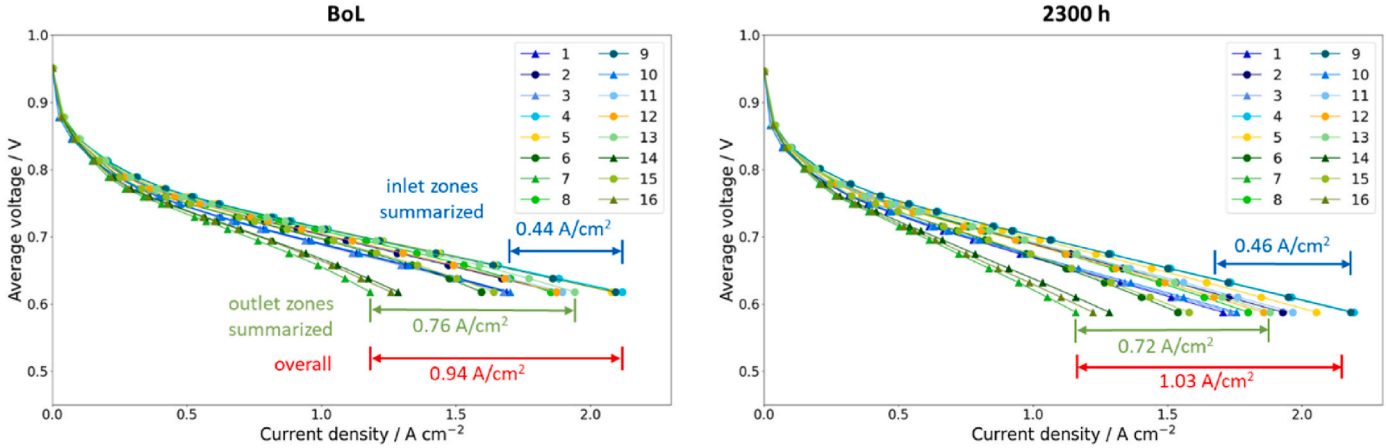


Fig. 7. Spatially resolved U–I curves plotted from the CDDM data for the 16 selected zones during the nominal RealHyFC polarization curves, at BoL (on the left), and 2300 h after air/air start (on the right). At each fixed voltage value, the local current density is averaged over all the CDDM segments of the zone. Arrows represent the current density span of a selected group of zones at 0.6 V.

maximum spread occurs both at BoL and after 2300 h between zones 4 and 7 (respectively at coolant and cathode outlet). The maximum spread is 0.94 A cm^{-2} at BoL and increases to 1.03 A cm^{-2} after 2300 h. An increase in the maximum difference can also be observed in the inlet areas. The spread increase in the inlet areas is of 20 mA cm^{-2}. In contrast to the inlet areas, the spread in the outlet areas decreases from 0.76 A cm^{-2} to 0.72 A cm^{-2}. These observations are consistent with the averaged distribution at 1.8 A cm^{-2} for the 10 first and 10 last segments in Fig. 6.

Fig. 8 shows how the maximum difference between zones 4 and 7 varies with the operating conditions, in the partial sensitivity test at BoL and after 2300 h at 1.8 A cm^{-2}. The variations characterize the individual operating conditions according to Table 2. The values for the operating conditions increase with increasing number of variations. At a pressure of 60 kPaG/50 kPaG, 1.8 A cm^{-2} could no longer be measured because the cell voltage fell below 500 mV, which was defined as the lower voltage limit for the durability test.

The gap in the averaged current density between the two zones decreases both at BoL and after 2300 h when increasing humidity, pressure, and stoichiometry. The opposite occurs when increasing temperature, showing that the inhomogeneity across the active area increases with temperature. A comparison of all the two curves at BoL and after 2300 h shows that the difference between the two zones is always greater after aging, hence showing an increase in the overall heterogeneity for all conditions. However, this difference in inhomogeneity decreases with increasing temperature showing that degradation has therefore a greater influence on the current density distribution at lower temperatures. With stoichiometry it is the same, aging has a greater influence on the current density distribution at low stoichiometry. With pressure and humidity, it is the reverse, the influence of aging on the current density difference increases with increasing pressure and increasing humidity, respectively.

3.3. Electrochemical impedance spectroscopy

This paper focuses on the results of the EIS measurements at 1.8 A cm^{-2}, where the mass transport resistance, usually comes into play in both the aging process and the sensitivity analysis. Furthermore, the spatially resolved measurements at 1.8 A cm^{-2} exhibit the lowest measurement noise. The results of the cell EIS measurement over time under nominal RealHyFC conditions are shown in Fig. 9. In order to complete the discussion about the processes involved, in addition the EIS results at a current density of 0.18 A cm^{-2} are also shown in Fig. 9a. It should be noted that EIS measurements at cell level represent an

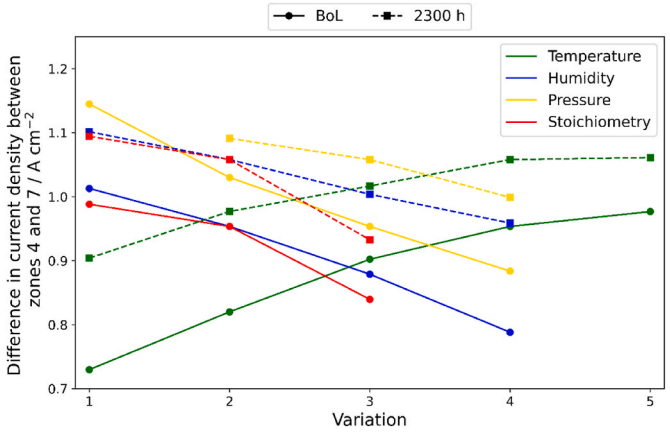


Fig. 8. Current density difference between zone 4 (highest current density) and zone 7 (lowest current density) at 1.8 A cm^{-2} over the different variations of the sensitivity test (cf. Table 2) with increasing values over the variations in all cases at BoL and after 2300 h.

average of the processes across the active area. In the measurements underlying this work, capacitive artifacts occur in the high frequency range. For this reason, the frequency range at cell level was limited to 100 mHz – 1180 Hz.

The averaged Nyquist plot of all 10 cells of the stack over time (Fig. 9 a) show for both load points a shift of the curves along the real axis by the value of the high frequency resistance (HFR). The HFR represents the ohmic losses, which comprising the membrane resistance as well as contributions from electronic contact resistances and electrode-related effects, including ionomer distribution and possible swelling phenomena. Nevertheless, according to Heinzmann et al. [40], the ohmic losses consist mainly of the ionic component, which can be attributed to proton transport in the membrane. In addition to the high-frequency intercept, intermediate- and low-frequency intercepts can be identified. The low frequency resistance (LFR) represents the sum of all losses. The sections between the intercepts LFR, IFR and HFR can be assigned to two processes. These are attributed to mass transport in the low frequency range and to the charge transfer of the oxygen reduction reaction (ORR) on the cathode side in the intermediate frequency region. At higher current densities, mass transport effects become increasingly dominant, leading to a clearer separation between intermediate- and low-frequency features. In contrast, at lower current densities, the impedance response is mainly governed by charge transfer processes, and mass transport contributions are less pronounced.

The three processes can be represented by an equivalent circuit diagram shown in Fig. 9 b). The resistance R0 represents the membrane resistance. The charge transfer of the ORR and the mass transport limitations can each be represented by an R-CPE element (constant phase element, CPE). Generally, the findings from the fitting of the equivalent circuit model support the qualitative discussion of the Nyquist diagram in the previous section. The Nyquist plot shows an overall increase of the impedance over time at 1.8 A cm^{-2}, with a clear gap between the spectra until 1150 h and the spectra recorded at 2300 h. The distribution of the fitted resistances is shown in Fig. 9 c). The mean value across all cells is shown with the standard deviation indicated in the form of error bars. The membrane resistance increases slightly over time, with a clearer jump after 1150 h. The increase in membrane resistance can be attributed to membrane degradation. Chemical aging of the membrane due to radical attacks reduces proton conductivity and increases membrane resistance [41–43]. In addition, an increase in interfacial contact resistances, e.g., between the gas diffusion layer and the bipolar plate, may also contribute to the observed rise in resistance. The ORR resistance initially decreases slightly and then increases again. The increase in charge transfer resistance is primarily attributable to catalyst aging. Typical mechanisms of catalyst layer aging reported in the literature include platin particle growth due to Ostwald ripening, dissolution and platinum agglomeration, as well as carbon corrosion. These processes lead to a reduction in the ECSA and thereby impair the oxygen reduction reaction [10,44,45]. The mass transport resistance behaves contrary to what is described in the literature. Over time, the mass transport resistance normally increases due to blocked gas diffusion pathways caused by degradation mechanisms such as liquid water accumulation, loss of hydrophobicity in the gas diffusion layer, pore structure changes in the catalyst layer, and carbon support corrosion [9,10,46]. However, the mean fitted values here remain practically constant over the timespan covered by the test reported here. This may be due to the specific test conditions avoiding high potentials or the test duration which might be not long enough to trigger the relevant processes in the GDL and catalyst layers. However, these trends are difficult to interpret. In summary, it can be stated that this aging test caused only minor effects in the average impedance measured at cell level.

The distribution of resistances across the active area and their change over time are also represented using spatially resolved EIS measurements. For this purpose, the spatially resolved EIS spectra were fitted using the same equivalent circuit diagram mentioned above. The frequency range of the spatially resolved measurements was limited in the

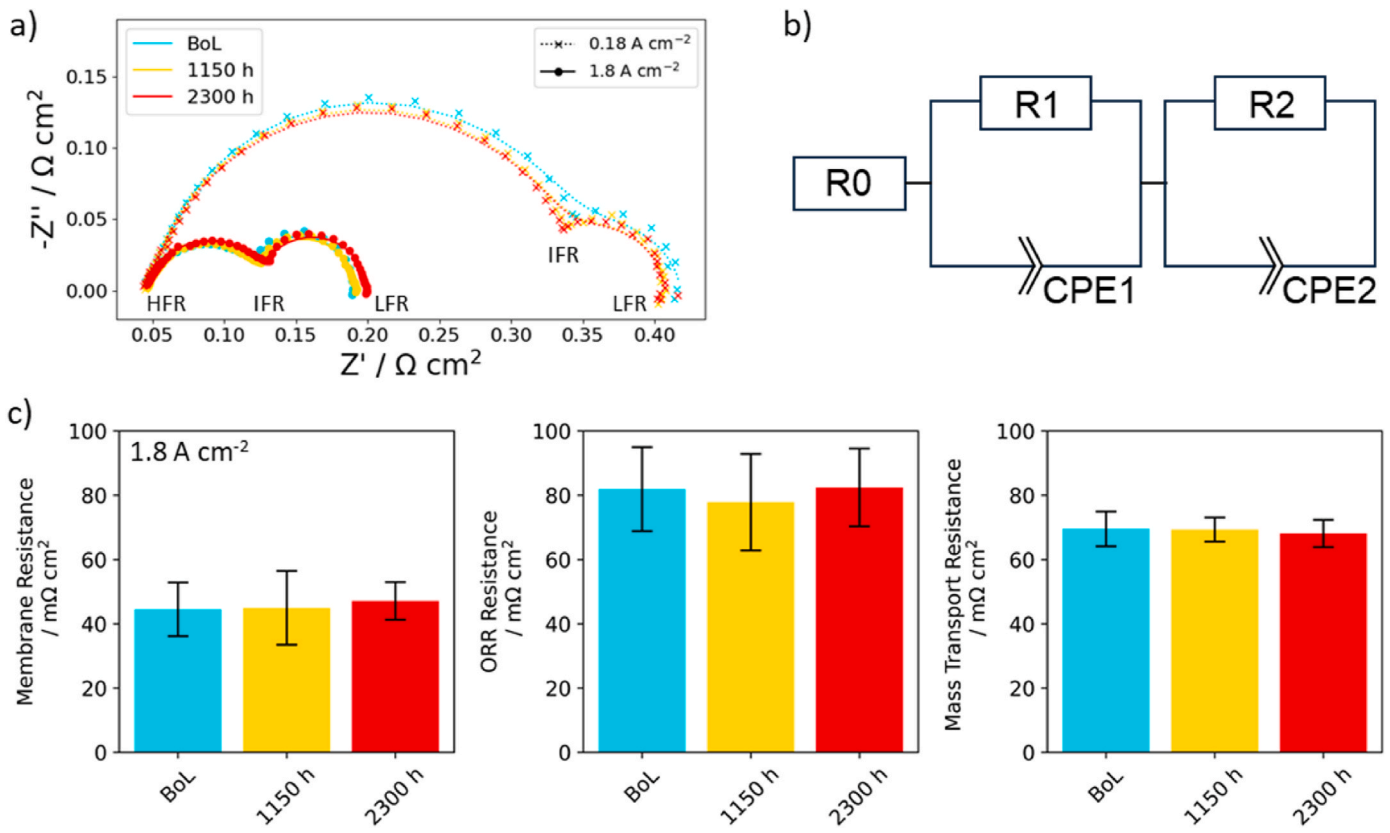


Fig. 9. Nyquist plot of mean cell impedance spectra from all 10 cells of the stack over aging for 0.18 A cm^{-2} and 1.8 A cm^{-2} with labeling of the high-frequency resistance (HFR), the intermediate-frequency resistance (IFR) and the low-frequency resistance (LFR) (a), equivalent circuit model (b), and fitted resistances for 1.8 A cm^{-2} averaged over all cells with marking of the standard deviation by error bars (c). R0: membrane resistance, R1: charge transfer resistance of the oxygen reduction reaction (ORR), R2: mass transport resistance, CPE: constant phase element.

upper frequency range due to measurement artifacts. For the cathode inlet zones 1, 3, 10 (cf. Fig. 1) a maximum frequency of 1180 Hz was set and for the remaining zones a higher frequency limit of 421 Hz was defined. To map the resistance values across the active area, the measured values of the individual zones were set at their zone centers. The values were linearly interpolated between the zones. For better resolution in the middle area of the active area, zones 5 and 12 were additionally mirrored across the flow direction. Zone 4 cannot be fitted in a meaningful and realistic way and was therefore excluded from the evaluation and replaced by the values from zone 9. Similarly, capacitive artifacts in the high frequency range occur in zone 7 that cannot be precisely localized, which is why the values from zone 16 were used for mapping the membrane resistance and ORR resistance for zone 7. The resulting resistances across the active area at BoL, 1150 h, and after 2300 h are shown in Fig. 10.

The mean values and standard deviation of the three resistances across the active area based on Fig. 10 are listed in Table 4. The membrane resistance increases from the center to the edges of the active area. The most obvious phenomenon is that the differences between low values (middle area) and high values are much more pronounced with aging. So, the distribution of membrane resistance across the active area becomes increasingly inhomogeneous with increasing aging, which is also indicated by the standard deviation. The membrane resistance increases in average by 9.75 % over the first 1150 h. The increase in membrane resistance over time observed in the boundary regions can, in agreement with the cell-level results, be attributed to both mechanical and chemical membrane degradation. In these regions, enhanced local stress conditions and inhomogeneous operating environments may accelerate degradation processes, leading to a reduction in proton conductivity and thus higher membrane resistance. In contrast, the slight decrease in membrane resistance observed in the central region may be associated with membrane thinning effects, which have been reported in the literature as a consequence of chemical degradation [8,10]. Such thinning reduces the effective transport path for protons and can therefore lead to a lower apparent membrane resistance, caused by ongoing material degradation. Of course, simultaneously the local cross over rate will increase, nevertheless this will not influence membrane resistance. Colombo et al. [8] have already described that the outlet region is most susceptible to membrane aging, a finding confirmed by the results presented here. The ORR charge transfer resistance generally increases from the cathode inlet to the cathode outlet, with again an accentuation of the local differences during aging. During the first 1150 h of aging, the resistance shows a decrease in the cathode inlet area and an increase in the cathode outlet area, resulting in a 37.2 % increase in the variation of the values. Of course, we are aware that the values are not randomly distributed over the active area, but follow clear physicochemical processes and boundary conditions. Nevertheless, statistical nomenclature can help to explain the increase in the variation. The increase of the ORR resistance in the outlet region is likely caused by enhanced catalyst degradation under locally harsher conditions, such as higher water content and reduced reactant availability. This is consistent with previous studies, which have shown that platinum degradation is more pronounced at the air outlet compared to the inlet, attributed to the higher local humidification [8]. In the following period of another 1150 h, the resistance increases in all areas, maintaining the previous shape of the distribution, thus a more homogeneous progression of catalyst degradation across the active area takes place. The mass transport resistance shows a strong increase from the cathode inlet to the cathode outlet, with the maximum being reached at the anode inlet. No significant changes over time are observed during the first 1150 h, then

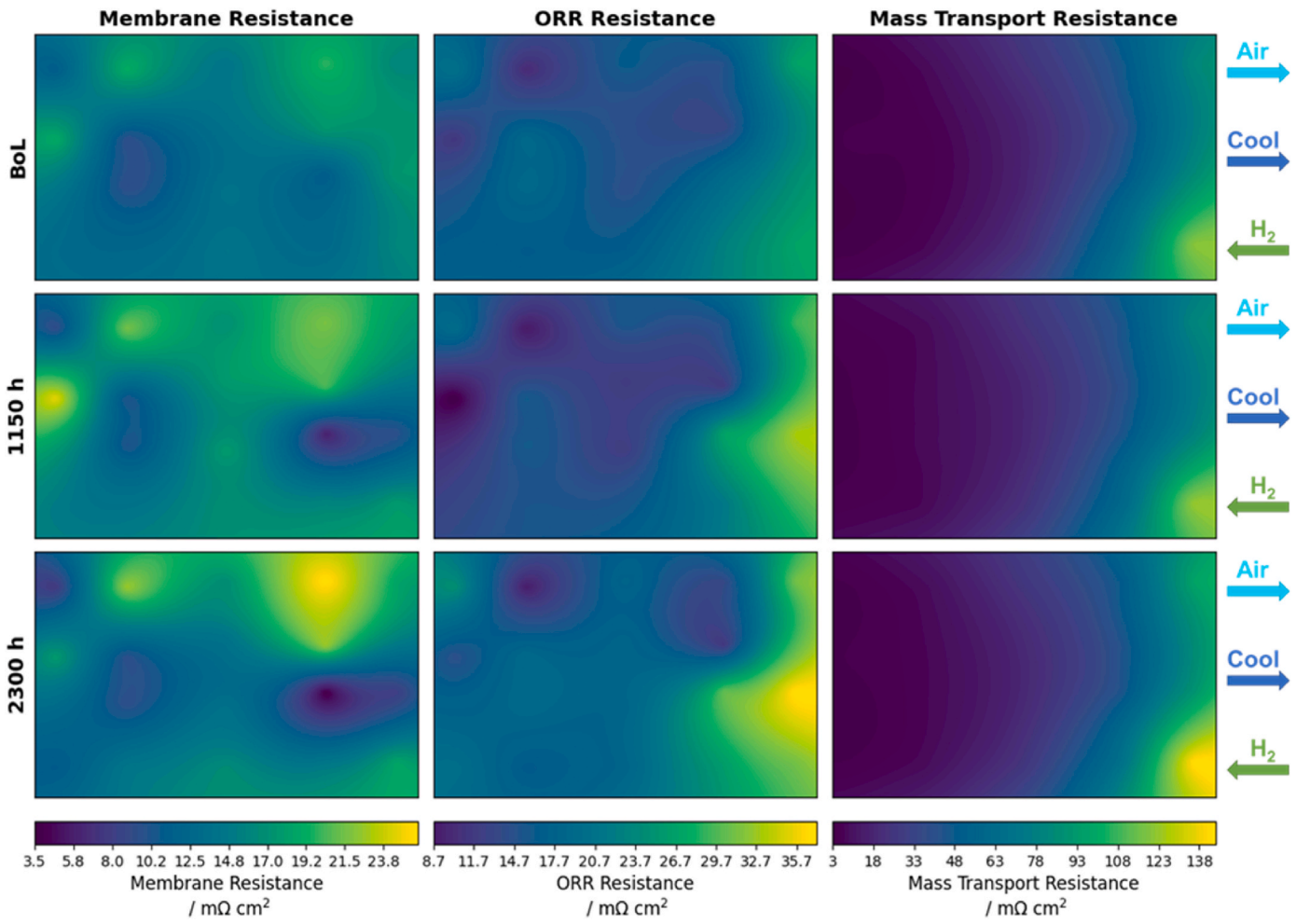


Fig. 10. Spatial distribution of resistances at 1.8 A cm^{-2} fitted from spatially resolved impedance spectra with linearly interpolated data between the zones over time. To increase the resolution in the middle area, zones 5 and 12 were mirrored along the flow direction. Left: membrane resistance, middle: ORR resistance, right: mass transport resistance, top: BoL, middle: after 1150 h and bottom: after 2300 h.

Table 4
Mean values and standard deviations of the fitted resistances plotted over the active area over aging from Fig. 10.

Time	Membrane Resistance /mΩ cm^2		ORR Resistance /mΩ cm^2		Mass Transport Resistance /mΩ cm^2	
	Mean value	Standard deviation	Mean value	Standard deviation	Mean value	Standard deviation
BoL	14.50	2.18	17.62	3.68	32.22	29.90
1150 h	15.91	2.99	17.17	5.94	32.34	28.78
2300 h	14.91	3.96	20.40	5.78	35.76	33.79

an 11 % increase compared to BoL is recorded, which is particularly evident both in the cathode outlet and the anode inlet area. This delayed increase suggests that structural changes in the porous media, such as degradation of the gas diffusion layer, loss of hydrophobicity, or pore blockage within the catalyst layer, progressively impair gas transport and become dominant at later stages of aging.

The sensitivity study provides information about the influence of operating conditions on impedance. Fig. 11 shows results of the partial sensitivity test at BoL and after 2300 h. The BoL data comes from the 8-cell short stack, and cell 5 which served as the reference for the spatially resolved measurements. The results after 2300 h are from the 10-cell stack that underwent the durability test, and the cell 6 used as the reference. As noted above, the identical MEA types and operating conditions were used in both stacks, ensuring comparability between the BoL and EoT investigations. In the following, the BoL and EoT sensitivity behavior of the two stacks will be investigated. The fitted Nyquist diagram of the pressure variation is shown in Fig. 11 a). After 2300 h the lowest pressure value could no longer be measured meaningfully. The fitted resistances associated with pressure variation are shown in Fig. 11 b). Fig. 11 also shows the fitted resistances for temperature (c) and stoichiometry variation (d).

The Nyquist plot (Fig. 11 a) shows that both the ORR resistance and the mass transport resistance exhibit a significant decrease with increasing pressure, both at BoL and after 2300 h. This is confirmed by the fitted resistance values. The observed decrease in impedance with increasing pressure is consistent with literature reports [47–49] and can be attributed to the enhanced reactant availability at the catalyst layer. The elevated oxygen concentration enhances the oxygen reduction reaction kinetics, thereby lowering the charge transfer resistance. In addition, the increased gas density improves diffusion and convective transport within the gas diffusion layer and catalyst layer, reducing concentration gradients and consequently diminishing mass transport resistance. In general, pressure has the greatest influence of the investigated operating conditions on these two resistances. This confirms the previously described results of the influence of operating conditions on cell voltage (see Fig. 5). The membrane resistance shows a slight decreasing trend with increasing pressure at BoL. This can be attributed to the increasing amount of liquid water as pressure rises, which leads to improved hydration of the membrane. At the same time, the increased

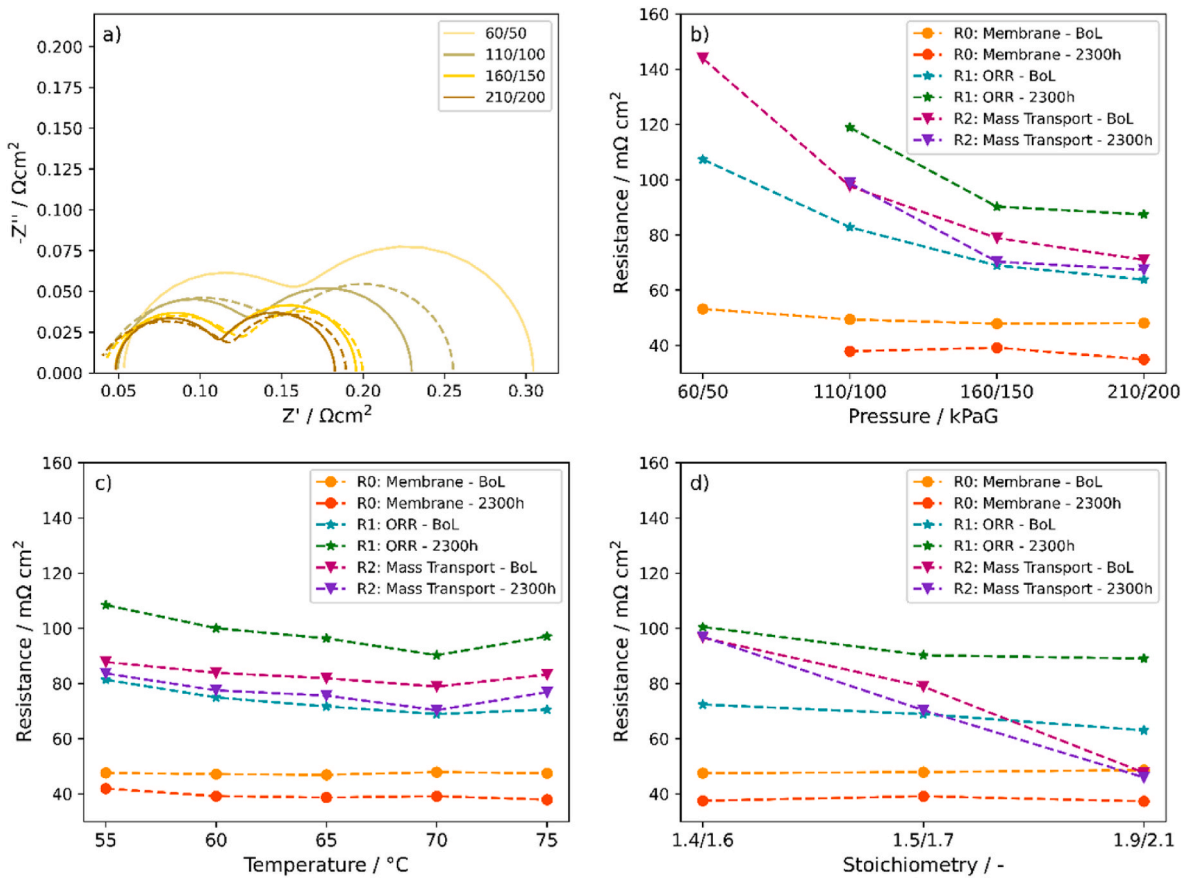


Fig. 11. Fitted Nyquist plot of the pressure variation (a) and fitted Membrane (R0) resistance, ORR (R1) resistance and Mass Transport (R2) resistance of the respective reference cell (cell 5 of 8-cell short stack for BoL and cell 6 of 10-cell short stack for 2300 h) for the pressure (b), temperature (c) and stoichiometry (d) variations at BoL and after 2300 h at 1.8 A cm^{-2}.

pressure reduces the flow velocities of the reaction media, thereby further reducing the dehydration of the membrane. After 2300 h, the membrane resistance remains stable.

For the temperature variation, the membrane resistance remains largely constant in both the initial state and the aged state. This is attributable to the fact that the relative humidity was kept constant across the entire temperature range, thereby ensuring uniform hydration of the membrane. The ORR resistance and mass transfer resistance show minimal resistances at 70 °C for both BoL and after 2300 h (Fig. 11 c). At higher and lower temperatures, the resistance increases in each case. We expect that the behavior of the ORR resistance and the mass transport resistance is based on the interaction of two opposing effects. An increase in temperature initially leads to an acceleration of the electrochemical reaction kinetics and transport processes, which would generally suggest a decrease in both resistances. However, this positive effect is evidently counteracted at higher temperatures. This may be due to increasing dehydration of the ionomer, which results in catalyst particles, which are only bound by thin ionomer films, being partially cut off from proton transport. The dehydration could be due to the non-linear increase of the saturation vapor pressure with temperature. Identical relative humidity in the gas channel may result in a lower local humidity at the catalyst layer at higher temperatures, thereby increasing the local resistance and slightly deteriorating ORR kinetics. In addition, higher temperatures may lead to increased water vapor release rates from the ionomer face, which would dynamically enhance the removal of water from the catalyst layer and further contribute to ionomer dehydration. Consequently, a smaller active surface area would be available for the reaction, while at the same time the effective transport paths for the reactants would increase. This dehydration may primarily affect the ionomer in the catalyst layer, which is in direct contact with the gas phase and may thus not immediately be reflected in the membrane resistance. At BoL, the aforementioned opposing effects largely compensate each other at 75 °C, so that only minor changes in ORR resistance from 70 °C to 75 °C are observed. After 2300 h of operation, degradation processes within the electrode structure probably enhance the effects of ionomer dehydration. As a result, the negative effects are no longer fully compensated by the improved kinetics, leading to a steeper increase as well as a greater overall variation in the ORR and mass transport resistance when deviating from the optimal temperature.

A change in stoichiometry has a significant influence on mass transport resistance. A decrease in stoichiometry leads to a reduction in the available oxygen, particularly in the cathode outlet area, which increases the mass transport resistance. This relationship has already been described several times in the literature [13,49]. The ORR resistance decreases both at BoL and after 2300 h with increasing stoichiometry and the membrane resistance stays stable for both cases.

The results of the sensitivity analysis illustrate the global trends and influencing factors at cell level. However, a spatially resolved analysis is necessary to gain a deeper understanding of the physical-electrochemical processes. The next section therefore analyzes the spatially resolved effects of the operating conditions, considering the average over the same 27 columns as for the current density distribution analysis. Fig. 12 shows the averaged resistance values along the cathode flow direction. As in Fig. 11, the data in Fig. 12 comes from two different stacks. The spatially resolved distributions of the resistances of the two stacks match very well and can therefore be compared with each other [27].

On the overall, between BoL and after 2300 h, the shapes of the plots

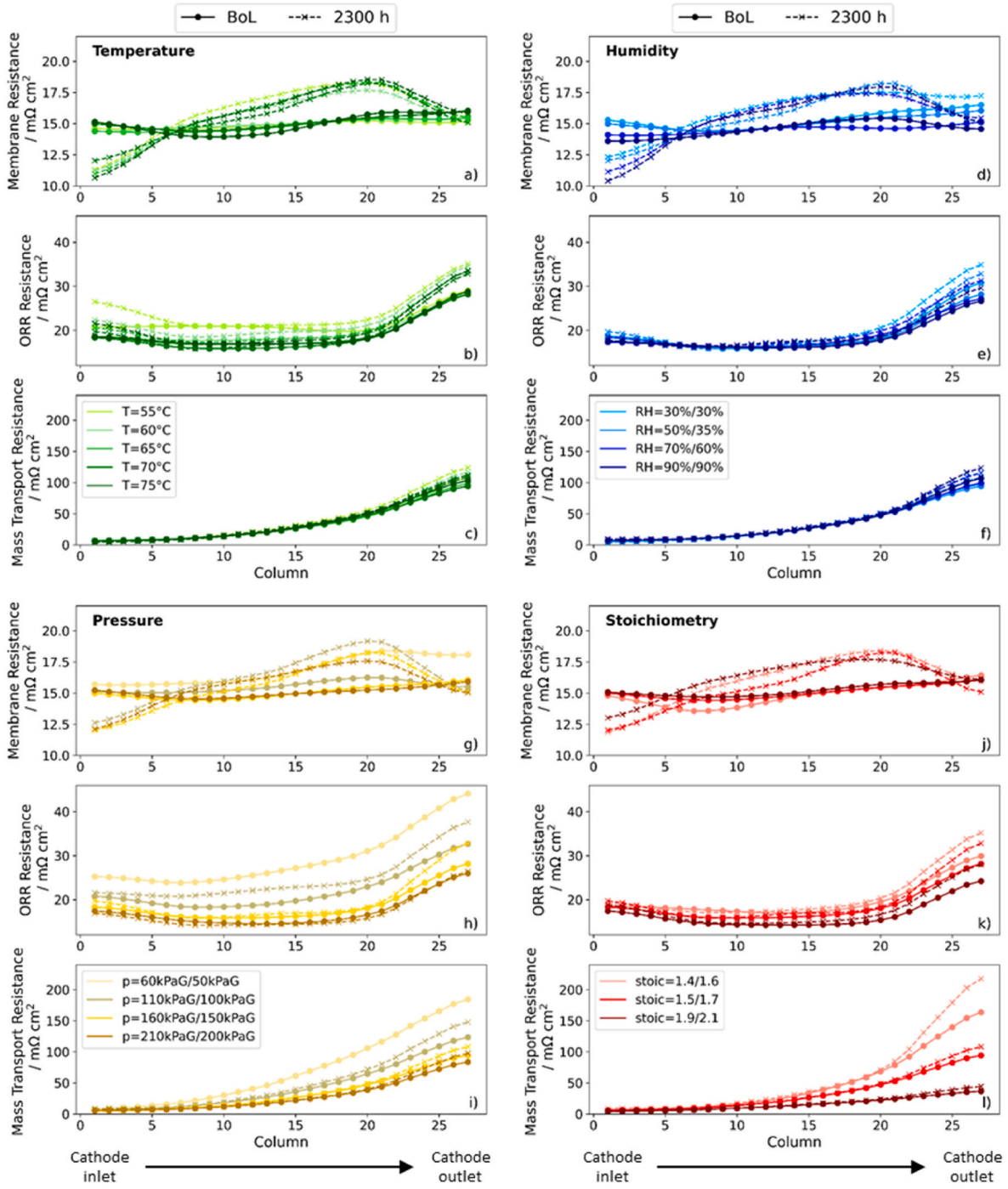


Fig. 12. Averaged resistances resulting from the spatially resolved EIS analysis within the partial sensitivity study plotted over the cathode flow field direction. Variation of temperature (a, b, c), humidity (d, e, f), pressure (g, h, i) and stoichiometry (j, k, l). In each variation, the membrane resistance (a, d, g, j), the ORR resistance (b, e, h, k), and the mass transport resistance (c, f, i, l) are shown.

are changing a lot for the membrane resistance, whereas they keep the same trend from inlet to outlet for the ORR charge transfer and the mass transport resistances. For the membrane resistance (a, d, g, j), the relatively flat profile at BoL changes with a strong decrease at the inlet, and an increase from segment 10 until a maximum appearing near segment 20. For the ORR charge transfer resistance, with aging, it increases even more near the outlet (b, e, k) but also all across the active area for the lower pressures (h). For the mass transport resistance, it also increases more with aging near the outlet but more strongly for both lower stoichiometries (l) and lower pressures (i).

The temperature variation in the membrane resistance shows that at BoL, the membrane resistance at the cathode inlet and outlet increases with rising temperature, but is lower in the middle. As the temperature decreases, the membrane resistance remains very stable at one value. After 2300 h, the membrane resistance at the cathode inlet decreases with increasing temperature, except for the 70 °C test, which shows the highest membrane resistance at the cathode inlet. The 70 °C line shows the greatest homogeneity across the cathode flow at 2300 h. The ORR resistance reflects the behavior at cell level. A temperature of 55 °C shows the highest resistances across the cathode flow. The 55 °C line stands out clearly from the others, especially at the inlet. At BoL, the temperatures above 55 °C at the cathode inlet are close together. In the

middle range, the ORR resistance decreases with increasing temperature up to 70 °C. Subsequently, the ORR resistance increases again. At the cathode outlet, all temperature curves are again very close together. After completing the load cycles, the temperature influence becomes greater and the curves are further apart. However, even after aging, the pattern remains that the ORR resistance decreases up to a temperature of 70 °C and then increases again. The same behavior is evident in both BoL and after 2300 h in the mass transport resistance. Here, the 70 °C curve shows the lowest resistance. Notably, mass transport resistance is not affected by aging in the inlet area, and the effects become greater towards the cathode outlet.

With regard to humidity, higher humidity has a positive effect on the membrane resistance [13,16,28]. This is particularly evident in the inlet area. At BoL and after 2300 h, there is a moderate variation between the different humidity curves in terms of lowest resistance. The ORR resistance shows a dependence on humidity and decreases with increasing humidity, as a result of improved proton conductivity with increasing humidity. The variation in humidity with aging shows an increase in ORR resistance, especially in the outlet area. The mass transport resistance is optimal at a humidity of 50 %/35 %. Higher humidity levels lead to an increase in mass transport resistance due to flooding effects, which is particularly evident in the cathode outlet area. Similarly, degradation only affects the outlet area.

The pressure variation influences the membrane resistance. As the pressure increases, the membrane resistance decreases. This can be attributed to the fact that as pressure increases, the media flow velocity decreases, thereby reducing the drying effect of the membrane. The ORR resistance is most influenced by the pressure under the operating condition variations considered. Here, a strong increase with decreasing pressure can be observed over the entire cathode flow, which intensifies towards the cathode outlet. When comparing the BoL with the aged curves, there is almost no difference at a pressure of 210 kPaG/200 kPaG. At the nominal pressure of 160 kPaG/150 kPaG, there is also hardly any difference in the inlet and middle areas. Only in the cathode outlet is a clear influence of aging effects visible. At a pressure of 110 kPaG/100 kPaG, the ORR resistance after 2300 h is consistently higher than at BoL. The mass transport resistance decreases with increasing pressure due to a higher oxygen partial pressure and consequently better oxygen supply to the catalyst layer. With regard to aging, mass transport resistance increases with increasing aging at all pressures. The increase is greater as the pressure decreases. The progression of ORR resistance and mass transport resistance confirm the results of the partial sensitivity test (chapter 3.1), which showed that the voltage loss with decreasing pressure is greater with increasing aging effects.

As the stoichiometry increases, the membrane dries out more and the membrane resistance increases. The ORR resistance shows a dependence on the stoichiometry, and the influence of the stoichiometry increases towards the cathode outlet. The ORR resistance decreases with increasing stoichiometry. This behavior can be attributed to the fact that increasing stoichiometry ensures a higher oxygen partial pressure and thus a better supply to the reaction sites, which facilitates the electrochemical reaction. The highest effect of aging in ORR resistance can again be seen in the outlet area. The mass transport resistance shows the greatest dependence on stoichiometry in relation to the other operating conditions considered. As with the ORR resistance, the influence increases towards the cathode outlet. Comparing BoL and EoT of the mass transport resistance, it becomes clear that aging has a significant influence on the outlet area and that there is no difference in the inlet area or in the middle area. Furthermore, the increase in mass transport resistance due to degradation effects increases with decreasing stoichiometry.

In general, the cathode outlet zone is strongly affected by the conditions having an effect of the oxygen feeding, and the limitations observed are even more pronounced after aging, probably because the aged catalyst layers are more sensitive to local issues like flooding.

4. Conclusion

In this work, a recently developed load cycle for heavy-duty applications was applied for the first time to a PEMFC stack with an active area of 280 cm^2. About 2300 operating hours of durability data were acquired, complemented by comprehensive electrochemical characterization methods to analyze both global and spatially resolved degradation phenomena. Polarization curves recorded at regular intervals reveal that performance degradation is most pronounced during the initial phase of the durability test and subsequently stabilizes. The sensitivity analysis of voltage losses indicates that aging effects are strongest at low pressures and elevated temperatures. Spatially resolved current density measurements, obtained using a CDDM, show a similar trend, with the largest changes occurring early in the test. BoL and EoT, the heterogeneity of the current density distribution decreases with increasing humidity, pressure, and stoichiometry, while higher temperatures lead to increased heterogeneity. Over time, overall heterogeneity increases, with stronger effects observed at higher pressure and humidity and at lower temperature and stoichiometry. Cell-level EIS measurements demonstrate an overall increase in impedance with progressing degradation. The EIS sensitivity analysis identifies optimal operating points at a temperature of 70 °C. Increasing pressure and stoichiometry generally reduce impedance. By extending the current density measurement system, spatially resolved EIS measurements across the active area were enabled. Fitting of the spatially resolved impedance spectra allowed the mapping of membrane resistance, cathodic ORR charge transfer resistance, and mass transport resistance, as well as the analysis of their evolution during aging. The membrane resistance increases predominantly in the boundary regions of the active area during the durability test, while a slight decrease is observed in the central region. In contrast, the aging of the ORR resistance in the first 1150 h is noticeable due to an increase in resistance in the outlet area. The mean value of the ORR resistance across the active area shows the greatest increase between 1150 h and 2300 h. A similar trend is observed for the mass transport resistance, which remains largely unchanged during the first 1150 h but increases significantly in the outlet region during the second half of the test. The sensitivity analysis shows that pressure has the strongest influence on ORR charge transfer resistance. At lower pressures, below 160/150 kPaG the durability test caused one of the major impacts observed with a strong increase in ORR charge transfer resistance. The mass transport resistance is most strongly affected by stoichiometry, with pronounced increases in the outlet region at low stoichiometric ratios. These stronger effects of aging are probably associated with the local degradation of the catalyst layers becoming more sensitive to the operating conditions. The catalyst degradation primarily affects the electrochemically active surface area and thus the charge transfer and mass transport resistances, while the ohmic resistance remains largely unaffected, as it is dominated by membrane and contact properties of the components including GDL and bipolar plate.

The results of this work provide fundamental insights that can be applied to system-level strategies for the heavy-duty sector. In particular, the spatially resolved analyses highlight the pronounced degradation phenomena in the air outlet region, underlining the need for optimized flow-field designs to improve local operating conditions. The significant increase in resistance at low stoichiometry and reduced pressure at the cathode outlet indicates that achieving the most homogeneous distribution of reactants possible should be a key objective of future flow-field and operating conditions optimizations. As a first step in this direction an optimization of the flow-field structures with regards to reactant homogeneity has been performed as part of the RealHyFC project, the results presented here are stemming from. The results of this optimization will be presented in a subsequent publication.

Furthermore, the pronounced sensitivity to operating parameters highlights the importance of precise control, enabling operating strategies to be specifically aligned with conditions that promote stability and durability. Overall, the results underscore the importance of

systematically taking spatial effects into account both in the design of stack hardware and in the development of suitable control strategies for heavy-duty PEMFC applications.

CRediT authorship contribution statement

Eva Fensterle: Conceptualization, Data curation, Investigation, Methodology, Validation, Visualization, Writing – original draft, Writing – review & editing. **Jürgen Hunger:** Investigation, Methodology, Resources. **Frank Rabsahl:** Investigation, Methodology. **Florian Wilhelm:** Methodology, Project administration, Supervision, Writing – review & editing. **Christian Kolle Christensen:** Methodology, Resources. **Adam Benson:** Methodology. **Rami Al-Khamissi:** Methodology. **Marine Tregaro:** Methodology, Writing – review & editing. **Sylvie Escribano:** Methodology, Project administration, Writing – review & editing. **Joachim Scholta:** Supervision, Writing – review & editing. **Markus Hölzle:** Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work was performed within the framework of the RealHyFC project. The RealHyFC project is supported by the Clean Hydrogen Partnership and its partners Hydrogen Europe and Hydrogen Europe Research under GA 101111904.

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